For polytopes with $\underline{d+3 \text{ vertices}}$, the complete classification becomes difficult, but not out of reach. Their affine Gale diagrams are configurations of signed points on a line. The main problem is to decide which different diagrams represent combinatorially equivalent polytopes. First results were due to Gale, while Perles developed the (Gale diagram) techniques necessary to analyze polytopes with $d+3$ vertices. The special case of simplicial polytopes was done in Grünbaum [252, Sect. 6.2]; the work was completed by Mani [376] and Kleinschmidt [333]. Explicit formulas for the number of d-polytopes with $d+3$ vertices were obtained by Perles [252] for the simplicial case and by Fusy [217] for the general case (correcting an error in an earlier solution by Lloyd [370]). Note that the octahedra that we considered before have $d+3$ vertices, for $d=3$.

The polytopes with $d+3$ vertices still do not have any unusual properties.

Finally, we arrive at the case of polytopes with $\underline{d+4 \text{ vertices}}$: here is the *threshold for counterexamples*, as Sturmfels [532] calls it. They can be analyzed in terms of planar point configurations — which can be arbitrarily complicated. In particular, for high enough d there are

- d-polytopes with $d+4$ vertices that do not have a realization with rational coordinates,
- d-polytopes with $d+4$ vertices for which the shape of a facet cannot be prescribed, and
- d-polytopes with $d+4$ vertices that have disconnected realization spaces.

We will now describe Gale diagram approaches to these three phenomena.

(a) A Nonrational 8-Polytope

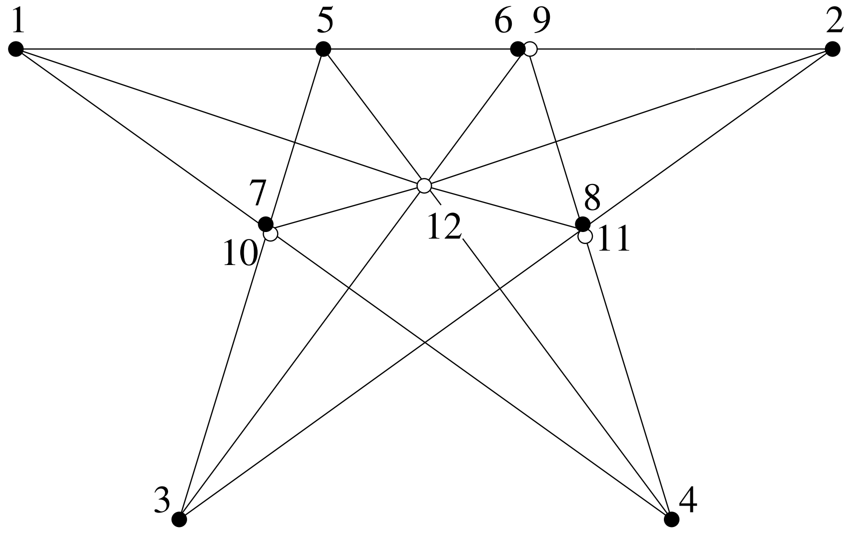
Using Gale diagrams, Perles has shown that there are nonrational polytopes, that is, polytopes for which there are no combinatorially equivalent polytopes with rational coordinates.

Example 6.21 (Perles). [252, p. 95] [96, Fig. 8.4.1]
There is a nonrational 8*-polytope with* 12 *vertices.*,

To see this, one verifies that the configuration G in the figure on the next page has three properties:

1. G cannot be realized with rational coordinates "as a matroid": there is no rational planar configuration of 12 points such that the same sets of points as in G coincide, respectively are collinear. (Essentially, there is a golden ratio involved in the construction of a regular pentagon, so it can only be realized with coordinates in a field containing $\sqrt{5}$.)

2. G is the Gale diagram of an 8-dimensional polytope with 12 vertices (check this!); a polytope represented by G cannot have all coordinates rational.



3. Consider any spanning configuration G' of 12 signed points in the plane (an affine Gale diagram in $\mathbb{R}^2$). If G' has the the same positive circuits as G, then the three pairs of points that coincide in G have to coincide in G' as well, and the triples and quadruples that are collinear in G have to be collinear in G' as well, because they are all positive vectors (unions of positive circuits).
   Thus also G' cannot be realized with rational coordinates.

Thus G is the Gale diagram of a nonrational polytope P. If P' is a polytope that is combinatorially equivalent to P, then its Gale diagram G' has the same positive circuits as the Gale diagram G, hence with part 3. above G' and P' cannot be rational either.

(b) Facets of 4-Polytopes Cannot be Prescribed

Perles apparently first observed that the shape of a facet of a d-polytope cannot in general prescribed; see Grünbaum [252, p. 96, Ex. 3]. Kleinschmidt [332] finally constructed a 4-polytope with 8 vertices for which the shape of a facet cannot be prescribed — this is the smallest dimension and the minimal number of vertices for such an example, because all facets can be prescribed for 3-polytopes, and for all d-polytope with at most $d+3$ vertices. With $d = 4$ and $n = 8$, Kleinschmidt's polytope can be constructed